OrderIQ — Day-by-Day Build Plan

One lecture a day. Learn it, then record it, then move on.

19 recording days + 2 finishing days · ~95 min of finished course

The daily rhythm (~90 min): learn and build the step yourself (40) → note what confused you (5) → dry run out loud, not recording (10) → record in one take (15) → rough cut and save the end-state SQL (15).

Two ordering rules: Lecture 1 is the finished-app demo, so it gets recorded *last* (Day 19). And every day ends with a saved `lecture_NN_end_state.sql` so you can jump back to any point without archaeology.

Part 1 — Foundations

	Lec	Video is about	What you learn / build that day
SETUP			
1	02	**What Snowflake is, and setting up your account**	Trial signup (Enterprise, region locked at signup). Create warehouse, database, three schemas, role, grants — all in SQL, not menus. Resource monitor for cost safety. Deliberately trigger a permission error so you can teach the "does not exist *or not authorized*" wording.
2	03	**The Snowflake mental model — storage, compute, credits**	Why storage and compute are separate and why that's the whole product. What a credit actually is. Micro-partitions at a high level. Warehouse sizing and auto-suspend economics. *Also today:* generate the synthetic dirty dataset (4 CSVs, ~150k order lines) so tomorrow has data.
DATA IN			
3	04	**Getting messy CSVs into Snowflake**	Internal stages, `PUT`, FILE FORMAT objects, `COPY INTO`, `ON_ERROR` behaviour. Break a load on purpose and read `COPY_HISTORY` to diagnose it. Four raw tables land, everything VARCHAR.
4	05	**Cleaning data is engineering, not chores**	Dedup with `QUALIFY ROW_NUMBER()`. `TRY_CAST` for the broken date formats. `NULLIF`. Decide and document the business rule for negative-quantity returns. Build the STAGING views.
5	06	**Designing a star schema**	Fact table grain (order line, not order — and why). Surrogate keys. Generating a date dimension. Build `FACT_ORDER_ITEMS` plus the three dims in ANALYTICS.
SQL ANALYTICS			
6	07	**The seven questions every business actually asks**	Revenue, AOV, units, top SKUs, category mix, city split, return rate — built as one reusable metrics view rather than seven throwaway queries.
7	08	**Window functions you'll really use**	Running totals, MoM and YoY with `LAG`, ranking within category using `QUALIFY`. Hit the classic off-by-one in period comparison and fix it on camera.
8	09	**RFM segmentation and cohort retention**	RFM scoring with `NTILE`. A monthly cohort retention grid. Practise reading the grid out loud — the interpretation is the value, not the SQL.
PYTHON			
9	10	**Connecting Python to Snowflake, safely**	Connector install, key-pair auth instead of hardcoded passwords, `.env`, `.gitignore` before the first commit. **End of Part 1 — package and upload it.**

Ship Part 1 on Day 9, don't wait. Ten lectures is a legitimate standalone product. Putting it live early gets you real buyers, real feedback, and the motivation to finish Part 2 — which is the actual risk here, not the recording.

Part 2 — The app and the AI

	Lec	Video is about	What you learn / build that day
PYTHON (cont.)			
10	11	**Snowpark, and why not pandas**	Run the same aggregation both ways and time them on screen. Pushdown compute vs pulling rows to your laptop. What breaks at 150 million rows.
11	12	**Structuring code like an engineer**	Refactor scattered queries into a `metrics.py` module with parameterised, injection-safe functions. Repo layout. Why notebooks alone don't get you hired.
12	13	**Building the dashboard**	Streamlit: KPI cards, trend chart, category breakdown, date and city filters. Caching so every filter change doesn't wake the warehouse.
GENAI			
13	14	**Why "just paste the schema into an LLM" fails**	Do exactly that, ask a real business question, get confidently wrong SQL. Diagnose why: no metric definitions, no business vocabulary, no synonyms. Collect real failure examples — you need them for this lecture to land.
14	15	**The semantic model**	Define tables, relationships, metrics and synonyms so "revenue", "sales" and "GMV" all resolve to the same thing. Add verified queries. Show the wizard, then show the YAML — the YAML is the skill.
15	16	**Chat with your data**	Call Cortex Analyst, get SQL back, run it, render chart plus written summary. Multi-turn follow-ups. Always expose the generated SQL so the user can audit it.
16	17	**Guardrails and evaluation**	Read-only role for the app, row limits, query timeouts. Then build a 15-question eval set, score it, and fix the semantic model where it fails. This is the most senior-looking artifact in the whole project.
SHIP			
17	18	**Deploy and automate**	Deploy as Streamlit in Snowflake for a shareable URL. A Task that refreshes the analytics layer daily. Final credit-spend check.
18	19	**Turning this into a job**	Push to GitHub. Record a 60-second demo *before the trial expires*. Exact resume bullets. The six interview questions this project attracts, and how to answer them.
19	01	**What you're building, and why it gets you hired**	Recorded last, sits first. Demo the finished app, then the architecture diagram, then honestly map each layer to the job-description bullets it answers. By now your on-camera voice is settled — this is why it's last.

Finishing days

	Task
20	Re-record Day 1's Lecture 2. It's your weakest take and it's the first thing buyers watch. Also re-shoot anything you flagged along the way.
21	Final edit pass, chapter markers, assemble the bundle (repo, dataset, semantic YAML, eval sheet, resume blurb, interview sheet), publish Part 2.

Things to watch across the 19 days

- **The trial clock.** 30 days from signup, then the account suspends and data is deleted about ten days later. Day 1 to Day 21 fits, but there's no slack for a two-week gap in the middle. If you do stall, expect to rebuild in a fresh account — which is exactly why the daily end-state SQL files matter.
- **Cortex daily cap.** Trial accounts without a payment method are limited to roughly ten credits a day of Cortex AI functions. Fine for this course, but don't loop hundreds of test questions on Days 13–16.
- **Preview features move.** Parts of the semantic-view and Cortex surface are still preview and the UI shifts. Narrate what's on your screen, and plan to re-record Days 14–16 within a year.
- **Runtime drift.** Log your actual runtime each day. If you're consistently at 7 minutes instead of 5, decide on Day 3 whether the course is 95 or 130 minutes — not on Day 15.
- **Don't re-record for stumbles.** Pause, re-say the sentence, cut it later. Restarting takes is what makes people abandon courses halfway.